

AI HACKATHON 2025

PROJECT PROPOSAL

PROJECT TITLE: KENYA SENTINEL - Bridging the Gap Between Citizens and the National Police through Smart Incident Reporting and AI-Driven Insights.

MEMBERS

1. McBride Obwoye – UI/UX Designer
2. Paul Muyali – UI/UX Designer
3. Charles Odari – Backend Engineer
4. Luke Mbogo – AI/ML Engineer
5. Agnes Kitua - QA & Deployment

ABSTRACT

Kenya Sentinel is an intelligent public safety platform designed to enhance real time collaboration between citizens and law enforcement agencies. The system empowers citizens to instantly report incidents through mobile and web interfaces, eliminating the need for physical visits to police stations. With integrated CCTV feeds, AI powered threat detection, and cloud based alerting, the system ensures that police receive verified, location based reports in seconds. The videos captured through the system can also be used as evidence in courts of law, reducing cases of insufficient or disputed evidence. Additionally, a smart news and awareness feature aggregates trending updates from credible sources to keep users informed on emerging security concerns and teach them how to act in case of a crime occurrence. By fusing technology, artificial intelligence, and community-driven reporting, Kenya Sentinel aims to build safer, faster, and more transparent communication channels between citizens and the National Police Service, redefining public safety in Kenya.

PROBLEM STATEMENT

Despite Kenya's ongoing efforts through laws such as the Prevention of Organized Crimes Act and recent crime-related amendments, the gap between citizens and law enforcement remains wide. Many crimes go unreported due to slow, manual reporting processes and limited communication channels. Citizens still rely on physically visiting police stations, causing delays in response and loss of crucial evidence. There is, therefore, a need for a digital solution that connects citizens and police in real time enabling quick incident reporting, CCTV based alerts, and public awareness to enhance community safety and trust.

OBJECTIVES

MAIN OBJECTIVE

By developing an intelligent mobile and web-based system (*Kenya Sentinel*) the project aims to bridge the gap between citizens and law enforcement through real time reporting, awareness dissemination, and AI driven surveillance and analysis.

SPECIFIC OBJECTIVES

1. By developing a citizen friendly platform that allows instant digital reporting of crimes, the system seeks to eliminate delays and promote quick response.
2. By integrating AI models with CCTV feeds, the system will automatically detect and record suspicious incidents and relay alerts to nearby police units.
3. By connecting citizens directly to the police through cloud technology, the system will improve accountability, transparency, and communication.
4. By sourcing and ranking trending security related news and updates from open APIs and trusted outlets, the system will keep the public informed and aware.
5. By incorporating safety awareness campaigns and educational content, the system will strengthen public participation in community policing.

JUSTIFICATION

Crime continues to pose a serious threat to the safety and well being of Kenyan citizens. Every day, people experience delayed responses to emergencies, underreporting of incidents, and limited access to accurate guidance on their rights. Current systems fail to provide timely, location specific information, leaving communities vulnerable and law enforcement struggling to act effectively.

Kenya Sentinel is a game changing solution. By offering a real time, centralized platform, it empowers citizens to report crimes instantly, receive urgent alerts, and learn how to protect themselves during emergencies. Simultaneously, it equips police and authorities with structured, actionable data to respond quickly and allocate resources efficiently. This system does not merely collect information it transforms it into actionable intelligence, bridging the critical gap between citizens and law enforcement.

Moreover, Kenya Sentinel is fully aligned with current legislation on crime prevention, digital communication, and data protection, ensuring legal compliance while fostering public trust. By combining technology, awareness, and policy, this project has the potential to significantly reduce crime impact, save lives, and strengthen community-police collaboration.

Investing in Kenya Sentinel is not just developing another app it is building a safer, smarter, and more responsive Kenya.

Implementation / Methodology

The Kenya Sentinel system will be implemented as a modern, scalable, and user friendly platform combining web and mobile applications with intelligent backend services. The

approach ensures real time crime reporting, alerts, and actionable insights for both citizens and law enforcement.

1. Frontend:

- **Web Application:** Built using React, providing an intuitive interface for citizens and police to report and monitor incidents. Features include dashboards, interactive maps, notifications, and educational content on citizens' rights.
- **Mobile Application:** Developed with React Native to ensure cross platform compatibility (Android and iOS). The mobile app mirrors the web functionality, allowing users to report crimes, receive instant alerts, and access guidance while on the move.

2. Backend:

- **Server & API:** Handles requests from both web and mobile clients, processes incoming data, and serves real time alerts.
- **Machine Learning Models:** Analyze crime data from reports, news feeds, CCTV videos, and integrated APIs to:
 - Alert the nearest police station automatically when an incident is reported.
 - Rank incidents by severity and location.
 - Detect trends or emerging hotspots.
 - Provide predictive insights for law enforcement and policymakers.
- **Database & Storage:** Securely stores user reports, crime data, and system logs. Implements encryption and role-based access control to comply with data protection laws.

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3. Integration & Workflow:

- Citizens submit reports via app or web → Backend processes and validates the data → ML models analyze severity and location → Alerts are sent to nearby users and police → Data is visualized on dashboards for law enforcement and policymakers.

4. Security & Compliance:

- Data privacy and integrity are prioritized with encryption, secure authentication, and adherence to Kenyan data protection regulations.
- Only authorized personnel can access sensitive crime reports.

This implementation strategy ensures that Kenya Sentinel is fast, reliable, and impactful, transforming crime reporting into a real time, actionable system that empowers citizens and strengthens community safety.

